2021 Winter Qual

III. $\Sigma = LL^T$: cholesky decomposition, L : full column rank

$y = Lz \sim N_n(0, \Sigma)$, $z \sim N(0, I)$

$y^T A y = z^T \underbrace{L^T A L}_{B} z$

$B^T = (L^T A L)^T = L^T A L = B$: symmetric

$LL^T A LL^T A LL^T = LL^T A LL^T$

Since L is full column rank, $\exists (L^T L)^{-1}$

$\cancel{(L^T L)^{-1} L^T L} L^T A LL^T A LL^T = \cancel{(L^T L)^{-1} L^T L} L^T A LL^T$

$\Rightarrow L^T A L L^T A L L^T = L^T A L L^T$

$\Rightarrow L^T A L L^T A L \cancel{L^T L (L^T L)^{-1}} = L^T A L L^T \cancel{L (L^T L)^{-1}}$

$\Rightarrow L^T A L L^T A L = L^T A L$

$\Rightarrow L^T A L$: idempotent

$\text{rank}(L^T A L) = tr(L^T A L) = tr(A L L^T) = tr(A\Sigma) = r$

$z^T B z - y^T A y \sim \chi^2_r$

$\blacksquare$

$\boxed{2}$

(a) $\mathbb{E}[y^T W y] = \mathbb{E}[\mathrm{tr}(y^T W y)]$

$$= \mathbb{E}[\mathrm{tr}(W y y^T)]$$

$$= \mathrm{tr}(\mathbb{E}[W y y^T])$$

$$= \mathrm{tr}(W(\Sigma + \mu\mu^T))$$

$$= \sigma^2 \mathrm{tr}(W) + \mathrm{tr}(\mu^T W \mu)$$

$$= \sigma^2 \mathrm{tr}(W) + \beta^T X^T W X \beta$$

$\boxtimes$

(b) $u(\beta, \lambda) = (y - X\beta)^T (y - X\beta) + \lambda^T (C\beta - b)$

$$\frac{\partial u}{\partial \lambda} = C\beta - b = 0 \Rightarrow C\hat{\beta}_n = 0$$

$$\frac{\partial u}{\partial \beta} = -2X^T(y - X\beta) + C^T\lambda$$

$$= -2X^T y + 2X^T X\beta + C^T\lambda = 0$$

$$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T y - \frac{1}{2}(X^T X)^{-1} C^T \lambda = 0$$

$$C\hat{\beta}_n = C(X^T X)^{-1} X^T y - \frac{1}{2} C(X^T X)^{-1} C^T \lambda = b$$

$$\lambda = 2[C(X^T X)^{-1} C^T]^{-1} (C(X^T X)^{-1} X^T y - b)$$

$$\hat{\beta}_n = \hat{\beta} - (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C\hat{\beta}$$

$$+ (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} b$$

$$= (I - B)\hat{\beta} + c, \text{ where } c = (X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} b$$

$\boxtimes$

(c) $\hat{\beta}_c = (I-B)\hat{\beta} + c$

$\mathrm{Cov}((I-B)\hat{\beta}+c,\ C\hat{\beta})$

$= (I-B)\,\mathrm{var}(\hat{\beta})\,C^T$

$= \sigma^2 (I-B)(X^TX)^{-1}C^T$

$= 0 \qquad \therefore\ \hat{\beta}_c \perp\!\!\!\perp C\hat{\beta}$

$\boxtimes$

(d) $\mathrm{var}(\hat{\beta}_c) = \sigma^2 (I-B)(X^TX)^{-1}(I-B)^T$

$= \sigma^2 \underbrace{(I-B)(X^TX)^{-1}}_{=Q}$

$Qx = 0 \Rightarrow x = 0$ invertible

$\Rightarrow x \neq 0$ not invertible

Let $y \neq 0$, $C \neq 0$ $\quad x := C^T y$

$\Rightarrow Qx = 0$ but $x \neq 0$ $\qquad \therefore$ rank-deficient.

$\boxtimes$

(e)

$$C\hat{\beta} - b = C(X^TX)^{-1}X^Ty - b$$

$$= C(X^TX)^{-1}X^T[y - X^TC(CC^T)^{-1}b]$$

$$(C\hat{\beta}-b)[C(X^TX)^{-1}C^T]^{-1}(C\hat{\beta}-b)$$

$$= [y - X^TC(CC^T)^{-1}b]^T \underbrace{X(X^TX)^{-1}C^T[C(X^TX)^{-1}C^T]^{-1}C(X^TX)^{-1}X^T}_{=A}[y - X^TC(CC^T)^{-1}b]$$

$$A^2 = A$$

$$\text{rank}(A) = tr(A) = tr(I_q) = q$$

$$\frac{1}{2}\mu^TA\mu = \frac{(C\hat{\beta}-b)^T[C(X^TX)^{-1}C^T]^{-1}(C\hat{\beta}-b)}{2\sigma^2} =: \lambda$$

$$\frac{(C\hat{\beta}-b)^T[C(X^TX)^{-1}C^T]^{-1}(C\hat{\beta}-b)}{\sigma^2} \sim \chi^2(q, \lambda)$$

$$\frac{SSE}{\sigma^2} = \frac{y^T(I-H_X)y}{\sigma^2}$$

$$= \frac{[y - X^TC(CC^T)^{-1}b]^T(I-H_X)[y - X^TC(CC^T)^{-1}b]}{\sigma^2}$$

$$\text{rank}(I-H_X) = tr(I-H_X) = n-p-1$$

$$\frac{1}{2}\mu^TA\mu = 0$$

$$\frac{SSE}{\sigma^2} \sim \chi^2(n-p-1)$$

$A(I-H) = 0$; independent

$$F = \frac{\chi^2(q,\lambda)/q}{\chi^2(n-p-1)/n-p-1} \sim F(q, n-p-1, \lambda)$$

$\boxtimes$

(f) $(C\hat{\beta}-b)^T[C(X^TX)^{-1}C^T]^{-1}(C\hat{\beta}-b) \leq q \cdot S^2 \cdot F_{q,n-p-1}(\alpha)$

$b \in \{b \in \mathbb{R}^q ; (C\hat{\beta}-b)^T[C(X^TX)^{-1}C^T]^{-1}(C\hat{\beta}-b) \leq q \cdot S^2 \cdot F_{q,n-p-1}(\alpha)\}$

$\Longleftrightarrow$ reject $H_0 : C\beta = b$

$\boxtimes$

$\boxed{3}$.

(a) Let $Q = A^TA(A^TA)^-$, $P = I$

$$QA^TA = A^TA, \quad PA^TA = A^TA$$

$$0 = (QA^TA - PA^TA)(Q^T - P^T)$$

$$= (QA^T - PA^T)(AQ^T - AP^T)$$

$$= B^TB \Rightarrow B = 0$$

$$AQ^T = AP^T$$

$$\Rightarrow A = A(A^TA)^-A^TA$$

(b) $AA^-A = A \Rightarrow A^T(A^-)^TA^T = A^T \Rightarrow (A^-)^T = (A^T)^-$

$$\Rightarrow (A(A^TA)^-A^T)^T = A((A^TA)^-)^TA^T$$

$$= A(A^TA)^-A^T \quad ; \text{symmetric}$$

$$\underbrace{A(A^TA)^-A^T}_{=A}A(A^TA)^-A^T = A(A^TA)^-A^T \quad ; \text{idempotent}$$

Let G_1 and G_2 s.t. $AG_1A^TA = A$,
$AG_2A^TA = A$

$$0 = (AG_1A^TA - AG_2A^TA)(G_1^TA^T - G_2^TA^T)$$

$$= (AG_1A^T - AG_2A^T)(AG_1A^T - AG_2A^T)^T$$

$$= B^TB \Rightarrow B = 0 \Rightarrow AG_1A^T = AG_2A^T; \text{ Invariant}$$